An Unbounded Entropy Cost of Localizing a Decoder

Jan Mikulík

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Abstract

We exhibit a family of classically local binary-output behaviors whose optimal centralized seed entropy is exactly two bits, while their optimal local seed entropy grows without bound. For $m = 2^k - 1$ settings per observer, the local optimum is exactly k bits. For arbitrary $m \geq 2$, equality in the lower bound $\log_2(m+1)$ is equivalent to the existence of a Hadamard matrix of order $m+1$.

The construction gives a concrete obstruction to extending a centralized causal Shannon bound to distributed decoders with a universal additive constant, even when all behaviors under consideration are locally realizable. Its entropy ingredient is established in the literature on pairwise independence. No historical priority or Lean certification is claimed.

1 A local behavior with a small centralized description

Fix an integer $m \geq 2$. Alice and Bob receive inputs $i, j \in \{1, \ldots, m\}$ and return bits $a, b \in \{0, 1\}$. Define the behavior

$$W_m(a, b \mid i, j) = \begin{cases} \frac{1}{2}\,\mathbf{1}_{\{a=b\}}, & i = j, \\ \frac{1}{4}, & i \neq j. \end{cases}$$

Thus equal inputs require a shared fair bit; unequal inputs require independent fair bits.

A centralized realization uses a discrete random seed S, independent of the inputs, and a deterministic decoder

$$(A, B) = F(S, i, j).$$

Let $C_{\text{central}}(W_m)$ be the infimum of $H(S)$ over exact centralized realizations.

A local realization requires deterministic functions

$$A = f_i(S), \qquad B = g_j(S),$$

with no communication and with S independent of the inputs. Let $C_{\text{local}}(W_m)$ be the corresponding infimum. All random inputs used by either decoder are included in S. All entropies are measured in bits.

The behavior is local for every m: one may take m independent fair bits and return the bit indexed by the observer's input.

2 Exact centralized cost and the envelope

Theorem 1. *For every $m \geq 2$,*

$$\boxed{C_{\text{central}}(W_m) = 2.}$$

When (i, j) is treated as one action and (a, b) as one output in a one-step controlled tree, the information-envelope mean is also exactly $L(W_m) = 2$.

Proof. Let R, T be independent fair bits. Use

$$(A, B) = \begin{cases} (R, R), & i = j, \\ (R, T), & i \neq j. \end{cases}$$

This realizes W_m with a two-bit seed.

For any fixed unequal inputs, the output is uniform on four points. Its entropy is two bits, and it is a deterministic function of the seed. Hence every centralized seed has entropy at least two bits.

For equal inputs, the transcript information is identically one bit. For unequal inputs, it is identically two bits. The lower envelope of their information CDFs is therefore the CDF of the constant two-bit information variable. Its mean is two. □

3 Local realizations and pairwise independence

Theorem 2. *The local seed cost is exactly*

$$C_{\mathrm{local}}(W_m) = \min H(X_1, \ldots, X_m),$$

where the minimum ranges over all joint laws of pairwise independent fair bits $X_1, \ldots, X_m$. The minimum is attained.

Proof. In any local realization, the diagonal conditions give

$$f_i(S) = g_i(S) \quad \text{almost surely}$$

for each i. Since there are finitely many indices, these equalities hold simultaneously outside one null set. Put $X_i = f_i(S)$.

Each X_i is fair. For $i \neq j$, the prescribed law of $(f_i(S), g_j(S))$ is uniform on four points, so X_i and X_j are independent. Since the vector X is a deterministic function of S,

$$H(S) \geq H(X).$$

Conversely, any such joint law gives a local realization: take $S = X$ and return X_i and X_j. This proves equality of the optimization problems.

The feasible joint laws form a nonempty closed subset of the probability simplex on $\{0, 1\}^m$, described by linear marginal constraints. This set is compact, and Shannon entropy is continuous on it. The minimum is therefore attained. □

Theorem 3. *For every $m \geq 2$,*

$$\boxed{C_{\mathrm{local}}(W_m) \geq \log_2(m + 1).}$$

More strongly, every atom of a feasible joint law of $(X_1, \ldots, X_m)$ has mass at most $1/(m + 1)$.

Proof. Define the random sign vector

$$V = (1, (-1)^{X_1}, \ldots, (-1)^{X_m})^\top.$$

Fairness and pairwise independence imply

$$\mathbb{E}[VV^\top] = I_{m+1}.$$

If $V = v$ has probability $w > 0$, then

$$I_{m+1} - wvv^\top = \mathbb{E}\Big[VV^\top \mathbf{1}_{\{V \neq v\}}\Big] \succeq 0.$$

Since $\|v\|^2 = m+1$, evaluating this quadratic form at v gives

$$0 \le (m+1) - w(m+1)^2.$$

Therefore $w \le 1/(m+1)$.

Every positive-mass outcome consequently has information at least $\log_2(m+1)$, so

$$H(X_1, \ldots, X_m) \ge \log_2(m+1).$$

Apply Theorem 2. □

4 An exact unbounded separation

Theorem 4. *Let $k \ge 2$ and $m = 2^k - 1$. Then*

$$\boxed{L(W_m) = C_{\text{central}}(W_m) = 2, \qquad C_{\text{local}}(W_m) = k.}$$

Proof. Only the local upper bound remains to be proved. Index the m settings by the nonzero vectors $u \in \mathbb{F}_2^k$. Let S be uniform on $\mathbb{F}_2^k$, and define

$$X_u = \langle u, S \rangle \pmod 2.$$

Each X_u is fair. Distinct nonzero vectors over $\mathbb{F}_2$ are linearly independent, so for $u \ne v$ the map

$$s \longmapsto (\langle u, s \rangle, \langle v, s \rangle)$$

is surjective onto $\mathbb{F}_2^2$ with equally sized fibers. Thus X_u, X_v are independent fair bits.

Both observers return the indexed bit, using the same seed of entropy $H(S) = k$. The lower bound is $\log_2(m+1) = k$. □

Corollary 5 (Failure of a universal local overhead)**.** *There is no finite universal constant K such that*

$$C_{\text{local}}(W) \le L(W) + K$$

for every locally realizable finite behavior, when $L(W)$ is the information-envelope mean obtained by treating the joint input as one centralized action. This remains false with binary outputs for each observer.

Proof. For the family in Theorem 4,

$$C_{\text{local}}(W_{2^k-1}) - L(W_{2^k-1}) = k - 2 \longrightarrow \infty.$$

□

5 Equality and Hadamard matrices

A Hadamard matrix of order N is a matrix $H \in \{-1, 1\}^{N \times N}$ satisfying $HH^\top = NI_N$.

Theorem 6. *For every $m \ge 2$,*

$$\boxed{C_{\text{local}}(W_m) = \log_2(m+1) \iff \textit{a Hadamard matrix of order } m+1 \textit{ exists.}}$$

Proof. Put $N = m + 1$. Suppose first that the local optimum equals $\log_2 N$. By Theorem 2, it is attained by a feasible joint distribution. Theorem 3 bounds every positive atom by $1/N$. Equality in

$$H(X) = \sum_x p_x \log_2(1/p_x) \geq \log_2 N$$

forces every positive atom to equal $1/N$. There are therefore exactly N distinct outcomes.

Form an $N \times N$ matrix whose rows are the corresponding vectors

$$(1, (-1)^{x_1}, \ldots, (-1)^{x_m}).$$

The covariance identity from the previous proof gives

$$\frac{1}{N} H^\top H = I_N.$$

Since H is square, this also implies $HH^\top = NI_N$. Thus H is a Hadamard matrix.

Conversely, suppose a Hadamard matrix of order N exists. Multiply each row by the sign of its first entry, making the first column identically one. Orthogonality is preserved. Choose a row uniformly, and interpret its remaining m signs as binary variables.

Orthogonality to the first column makes each bit fair. Orthogonality between distinct remaining columns makes their sign product have mean zero. Together these facts imply that each pair of bits is uniform on four points. The uniform row index is therefore a local seed of entropy $\log_2 N$. The lower bound proves equality. □

6 Interpretation and attribution

The separation is entirely classical. Every behavior in the construction is locally realizable; no Bell violation or impossibility of local simulation is involved.

The centralized decoder can use the relation between both inputs. A local decoder cannot see the remote input. Consequently the local seed must support an entire family of compatible counterfactual response bits, constrained here to be pairwise independent.

This shows why the centralized envelope cannot, by itself, control distributed seed cost up to a universal constant. The missing structure is the restriction on which inputs each decoder may access.

The entropy lower bound for pairwise independent fair bits and its connection with Hadamard constructions are established results; see Gavinsky and Pudlák [1]. The present note expresses their consequence as a separation between centralized and local causal seed costs. It does not claim historical priority, resolve the Hadamard conjecture, or provide a Lean certificate.

References

[1] D. Gavinsky and P. Pudlák, "On the joint entropy of d-wise-independent variables," Commentationes Mathematicae Universitatis Carolinae **57**(3), 333–343 (2016). `https://doi.org/10.14712/1213-7243.2015.169`

[2] J. Mikulík, *One Seed for Every Adaptive Policy.* `https://github.com/jendamikulik/finite-Shannon-theorem`